

Grounding Mathematics Learning in Culturally Authentic Movement Practices: From Attentional Anchors in Balinese Dancing to Auxiliary Lines in Geometry Reasoning

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Abstract: Solving geometry problems often requires generating auxiliary lines. Yet, learning to generate auxiliary lines is challenging, because it involves envisioning figural elements that by definition are not yet overtly available for students' visual sensing or teachers' direct indexing. Espousing an enactivist perspective, this design-based research project explores dance as a culturally authentic grounding context for geometrical reasoning. Learning new dance movements often elicits the spontaneous mental generation of *attentional anchors*, imaginary perceptual structures enabling dancers to perform complex coordinations. We conjectured that, given appropriate semiotic resources, dancers could objectify their tacit attentional anchors from dance practice as explicit auxiliary lines for geometry practice. We report on a pilot evaluation of GRiD, a low-cost resource for blending dance and geometry. Microgenetic analysis of a young Balinese dancer's multimodal actions and utterances reveals the emergence of an interdisciplinary nexus of practice imbricating dance and mathematics and benefiting both discourses.

Introduction

We report on emerging results from an ongoing design-based doctoral research project that is taking on several research problems, two at the level of a pedagogical design framework (Abrahamson et al., 2020) and one at the level of building activities targeting specific disciplinary subject matter:

- *Framework-level research problems:* (1) How might the heuristic principles of embodied design, heretofore implemented using advanced digital technologies in industrialized countries, impact the STEM learning of remote students with low/no access to interactive electronic platforms? (2) How might the heuristic principles of embodied design, heretofore implemented in disciplinary contexts, such as mathematics classrooms, be situated in cultural–historically authentic practices?
- *Activity-level research problem:* In creating an embodied design for geometric reasoning, what evolutionary capacities might we draw on? How would we elicit those capacities? How would those capacities mobilize grounded mathematical discourse of the pedagogically targeted content?

In the remainder of the introduction, we expand on these three problems to explain our motivation for engaging in this line of inquiry. Then the later sections of the paper present our solution to the activity-level research problem as offering proof-of-concept solutions to the two framework-level research problems.

Embodied design: A need for low-cost and culturally situated framework expansions

Embodied design is an evolving research paradigm integrating embodiment *theory*, HCI *technology* innovation, and multimodal mixed *methods* to promote and investigate phenomena of teaching, learning, and designing (Abrahamson et al., 2020). Scholars of embodiment have challenged the traditional mind-body dualist perspective, i.e., the “cognitivist” assumption that the mind is epistemically separate from the body (Shapiro & Stolz, 2019). Instead, “4E” scholars—proponents of the notion that the mind is embodied, embedded, extended, and enacted—claim that the body, as well as its ecological niche, are inherent to, even constitutive of the cognitive system. Therefore, from an embodied cognition perspective, understanding and reasoning about the world necessarily begins from experiences of acting on the world (Shapiro & Stolz, 2019). Through these actions, per embodiment ontology, mathematical entities come forth as phenomenologically present objects we grasp, transform, and transport (Abrahamson et al., 2020).

The theory of embodied cognition can be applied to pedagogical practice by designing conceptually meaningful sensorimotor problems whose perceptual solutions ground the target content (Abrahamson, 2014). Some examples specifically of the *action-based genre* of embodied design-based research work in the field of teaching and learning include: (1) Mathematics Imagery Trainer activity tasks that challenge students to enact a target movement that maintains invariant some figural, relational, or quantitative property they thus first apprehend (Abrahamson, 2014); (2) Enacting Euclidean Transformation, where students learn translations,

reflections, and rotation through their collaborative action on a tarped Cartesian coordinate system (Soto, 2022); and (3) Balance Board Math, where vestibular sensory-seeking participants rock on a balance board to generate graphs with specific targeted properties (Tancredi et al., 2021). These projects all sought to create opportunities for students to first encounter concepts through bodily engagement with the environment and only later to signify these experiences using disciplinary frames of reference (see also Palatnik et al., 2023).

With an eye on equity, a key mission of embodied design is to enhance the accessibility of learning by embracing the variety of students' sensory, motor, and neural capacities as well as their diverse cultural epistemologies (Tancredi et al., 2021). In designing an educational product, embodied designers seek to develop a situated environment that invites students to engage intuitively in attempting to perform a motor-control task. The task then becomes increasingly challenging, stimulating students to shift their perception of the situation so as to enable them to move in a new way; these new perceptions constitute epistemic pivots in the sense that articulating these perceptual orientations multimodally with available semiotic resources engages students in conceptual discourse (Abrahamson & Sánchez-García, 2016). When students reason about their different movement solution strategies, they coordinate new conceptual insight (Abrahamson et al., 2020).

Advanced technology, such as augmented or virtual reality (XR), bears affordances for actualizing embodied design, because these platforms' immersive affordances can enhance students' sensorimotor experiences (Johnson-Glenberg, 2022). Yet whereas these digital solutions benefit students in resourceful environments, they are liable to increase educational gaps for students in disadvantaged socio-economic circumstances (OECD, 2022). As embodied designers, can we address the need to create low-cost technological educational artifacts that foster embodied learning despite digital scarcity? Consider the following anthropological perspective.

An important intellectual source for the theory of embodied design comes from the work of Reed and Bril (1996) on infants' early motor development. They coined the phrase *field of promoted action* to capture the practice of adults' explicit physical intervention, within a dedicated physical space, that occasions opportunities for young children to develop coordinative capacities enabling the prospective performance of culturally valued skills. For example, a Bambara mother in Sub-Saharan Africa holds her infant upside-down by the ankle to scaffold its coordination of lower- and upper-body skeletomuscular units, fostering physical power and agility. The researchers found that infants and toddlers raised in fields of promoted action located in remote rural villages advanced significantly in their motor capacities as compared to their urban peers (Reed & Bril, 1996).

Embodied-design draws on the cultural precedent of fields of promoted action as a theoretical justification for engineering digital architectures where children solve assigned motor-control problems to develop a perceptual grounding for the mathematical concepts they are later to understand (Abrahamson & Trninic, 2015). Still, what might action-based embodied design be in non-digital environments? Moreover, whereas embodied design has thus drawn on a culturally authentic practice to configure its *general* architecture, the framework ultimately *implements* its activities within a "cultural void," in the sense that students participate in the activities outside of any a priori meaningful context (Ba & Abrahamson, 2021). We speculated that presenting mathematics as augmenting *existing* cultural–historical practices could potentially ground the target skills more robustly within contextually authentic and personally relevant intergenerational discourse. Hence, we sought to create a field of promoted action where the geometric notion of auxiliary lines would spring forth as a spontaneous elaboration on existing praxis.

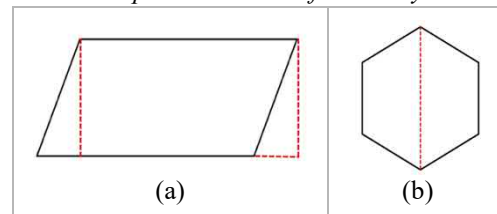
Geometry learning: Toward an embodied design for grounding auxiliary lines

A sweeping goal of all stakeholders in mathematics education is for students to understand the content they are learning, not just follow rote procedures (Skemp, 1976). In other words, the hope is for students to engage in the core mathematical practice of *reasoning*. In mainstream education, mathematical reasoning is fostered through various classroom activities that require students to make generalizations and support claims regarding logical–quantitative features of situations, such as the patterns or composition of spatially deployed objects (Stylianides, 2008). Yet, these features of the situations in question, what the reasoning will be *about*, are not immediately obvious or available—they emerge for the learner through exploration, investigation, conjecture, and evaluation (Radford, 2003). In so doing, students tune their perceptual attention toward the situations so as to mentally construct them in ways conducive to solving the problem at hand (Abrahamson & Trninic, 2015). Consider, for example, how the relation between geometrical shapes could be perceived in Figure 1.

The dashed red lines in Figure 1 serve as auxiliary lines. *Auxiliary lines* are supplemental diagrammatic elements that a geometry problem solver introduces onto the shapes in question to facilitate the discernment of implicit figural properties and latent relations key to constructing a mathematical proof (Palatnik & Sigler, 2018). For example, auxiliary lines can help us appreciate the structural premise for calculating the area of a

parallelogram by instead calculating the area of a rectangle sharing the same base and height (Figure 1a) or to realize that a hexagon is composed of two trapezoids (Figure 1b).

Figure 1
The Conceptual Function of Auxiliary Lines



The educational goal of geometric reasoning, however, is for problem solvers to figure out *themselves* what supplementary markings would mobilize their geometric argumentation. And so, students eventually receive images without ready-made auxiliary lines. Yet, it is precisely this first stage of generating useful auxiliary lines *ab nihilo* (i.e., “out of nothing”) that students find difficult: We are asking students to attend to something that is not there, at least, not as a bonafide sensory stimulus. In other words, *we are asking students to perceive something they cannot see*. As such, we face a double referential and pedagogical conundrum: a teacher cannot, in principle, point to an auxiliary line that isn’t there, and, in any case, the teacher may prefer not to gesture (depict, trace, etc.) that line either, so as not to disclose its form, location, or logical affiliation to the entire display. How, thus, might these lines come forth for students? How might students learn to generate these lines themselves?

In seeking to create a design for auxiliary lines, we join scholars who advocate for geometry education that draws on students’ intuitive capacities (Doorman et al., 2020). Yet, what naturalistic capacity might ground the construction of auxiliary lines?

- The first clue is found in evidence that dancers spontaneously generate imaginary figural elements to support the enactment of a dance movement (Clark & Ando, 2014). More broadly, Hutto and Sánchez-García (2015), who draw on the enactivist paradigm, coined the phrase *attentional anchors* to characterize perceptual resources, often involving imaginary constructions, that tighten one’s grip on the domain of action. Indeed, when embodied design-based researchers posed students with motor-control problems, the students spontaneously generated attentional anchors to enable the enactment of solution movements (Abrahamson & Sánchez-García, 2016).
- Moreover, Abrahamson and Shulman (2019) explicitly compare mathematics and dance pedagogy to argue for the shared cognitive function of attentional anchors in solving movement problems across both disciplines. As such, design-based researchers have theoretical grounds for situating mathematics learning within dance practice.
- Further, Gerofsky (2011) found that students who experience “being” a concept through kinesthetically enacting it in whole-body mimetic movement are generally more creative in problem-solving than their peers who report “seeing” the concept as a static display that is separate from their body.
- Furthermore, movement practitioners can represent tacit aspects of their performance skill using externally inspectable semiotic resources (Kirsh, 2010). As such, by overtly marking their covert perceptual operations, as they solve a movement task, students could construct external representations that serve as shareable objects of thought, thus supporting inter- and intra-personal communication.
- Finally, design-based researchers have characterized unique pedagogical affordances of *collaborative* body-immersive activities for geometry learning (Palatnik et al., 2023; Vogelstein et al., 2019). Yet, whereas these geometry designs have demonstrated the spontaneous emergence of mathematical problem-solving, imagination, and discourse from students’ situated task-based interactions with educational resources, the studies did not address the enactivist grounding of these cognitive activities. Namely, can we design geometry activities where vital aspects of formal disciplinary practice, such as building auxiliary constructions, emerge from students’ spontaneous solutions to problems of sensorimotor engagement?

Considering these insights, we conjectured that *students’ evolutionary capacity to generate attentional anchors as perceptual solutions to movement problems in dance practice could be tapped as a means of training them to construct auxiliary lines as solutions to geometry problems in mathematical practice*. Drawing on principles of action-based embodied design (Abrahamson, 2014) and the semiotic-cultural theory of objectification (Radford, 2003), the present project therefore conceptualizes the emergence of geometry discourse

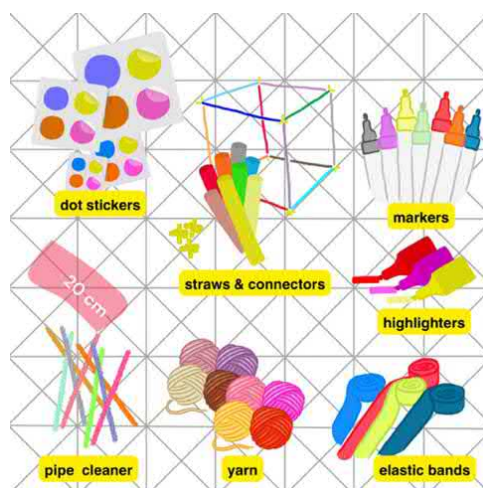
(i.e., discussing auxiliary constructions) as students' semiotic objectification of their situated perceptual solutions to motor problems of enacting goal movements (i.e., their attentional anchors).

In summary, our design-based research seeks to situate embodied activities for geometry learning within culturally authentic dance practices, using low-cost dedicated materials. We conjectured that, given appropriate semiotic resources, the attentional anchors students spontaneously generate in learning to move in new ways (i.e., in solving motor-control problems) could be objectified as auxiliary lines serving in solving geometry problems. As such, we are attempting to ground mathematical reasoning within dance practice.

Geometry resources in dance (GRiD): Overview of a design solution for grounding mathematical reasoning in movement practice

GRiD (see Figure 2) comprises a gridded mat, markers, stickers, highlighters, straws, connectors, yarn, pipe cleaners, and elastic bands (graphic artwork: Made Ariawan). Students were invited to utilize the GRiD to solve our designed problems (for additional dancing-geometry GRiD activities, see Apsari & Abrahamson, 2024). In so doing, the researcher encouraged the students to utilize a variety of multimedia and multimodal semiotic resources, including actual enactment, pantomime, gestures, speech, text, and drawings.

Figure 2
GRiD's Material Components



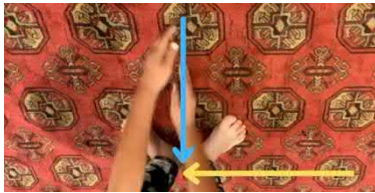
Implementation and evaluation of GRiD: A case study

We implemented the design with 77 fifth-grade students in Bali and California (9-11 years old) who joined individual or group sessions in a 60-minute teaching experiment (Steffe & Thompson, 2000) with the first author. The audio-video data collected during the study were analyzed qualitatively using micro-genetic ethnography (Siegler, 2006) and elements of the ethnomethodological approach to conversation analysis (Mondada, 2019) to take into account multimodal semiotic resources that participants presented during the problem-solving activity. In this paper, we will present selected results and findings of the activity with Anna (pseudonym), a 10-year-old Balinese dancer in training, that we argue, exemplifies the pedagogical affordances of our. Specifically, we contend, GRiD offers students engaged in solving sensorimotor coordination problems opportunities to objectify their tacit perceptual operations in disciplinarily normative forms, thus grounding geometry discourse in dance practice.

The researcher guided Anna (**A**) to practice a dance posture called Tapak Sirang Pada (henceforth, TSP), where the heels touch while the feet are rotated out, creating a 90° angle. The notion of a right angle as well as its standard Mesopotamian measure as 90° has entered Balinese culture and vernacular, so that some modern Balinese dance teachers might casually refer to the TSP aperture in those terms. However, little if anything is ever said regarding how the students might assess the right-angled quality of their own postural form. Soon, Anna demonstrated how she usually performed and evaluated her posture (see Transcript 1). At this moment, GRiD has not yet been introduced—a carpet at the research site happened to feature various traditional forms, but these were never alluded to. It is important to appreciate that Anna was not aware of the researcher's mathematical "ulterior motives"—all geometric discourse manifest in these excerpts was entirely spontaneous.

Transcript 1

Justifying the Angle of Tapak Sirang Pada without GRiD

 (a)	 (b)	 (c)
A: If... this section [blue arrow traces her bimanual gesture], and this section [yellow arrow traces her gesture] would be a right angle,	A: this would be like the 45 degrees [gray arrow traces her back and forth RH gestures, the dashed blue line traces her earlier vertical gesture as in Image 1a],	A: and then that would be the 45 degrees.

When Anna performed TSP, she perceived her feet as enclosing a 90° angle. To communicate this covert perception, Anna engaged in a make-shift geometrical argumentation. First, she constructed auxiliary lines that divided her stance into two spatial regions, each comprising a right angle (Transcript 1a-b focuses on the right side; by eliding an analogous argumentation for the left side, Anna conversationally implies that the same rationale obtains on the left). On each side, Anna submitted, her foot constitutes an angle bisector that halves each of the right angles, respectively, resulting in two 45° angles, on each side (Transcript 1b-c). From these lemmas, Anna concluded that the angle measurement between her feet sums up to a 90° angle. Q.E.D.

In response to Anna's explanation, the researcher placed the GRiD on the floor, framing it for Anna as a medium that could possibly help her to further demonstrate, perhaps with greater detail and clarity, these imaginary linear structures she is citing in Transcript 1. As the transcript makes evident, Anna immediately attended to the mat's lines and their intersections, apparently adjusting her body position so that her covert imaginary lines (Transcript 1a) align with the overt grid lines (Transcript 2).

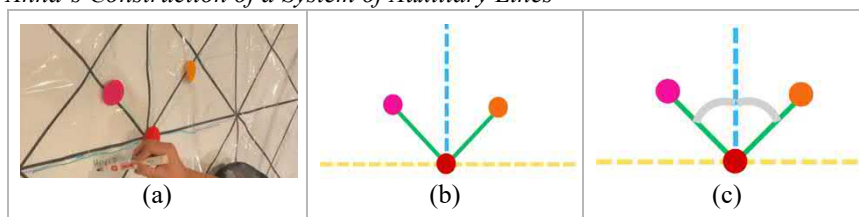
Transcript 2

Making the Invisible Lines Visible

 (a)	 (b)	 (c)
A: Like with my two heels attached to each other would be this big... [circular red arrow traces gesture direction]. And then, mmm, like what I said earlier,	A: I will have my horizontal line from ... mmm what it— from my heels, and it will just go horizontal [gestures away from the heels with both hands simultaneously; yellow arrows trace gestures' directions],	A: and my vertical line [blue arrow traces gesture direction] will be right here and go right in between my heels like right there.

Anna justified the positioning of her feet by focusing on the angle's vertex, where her heels met. Using GRiD, she first gestured and drew the imaginary points and lines that helped her determine she had performed TSP correctly. Now that the lines and the angle vertex feature explicitly on GRiD, Anna could elaborate on how the auxiliary elements relate spatially both to the body and to each other—she further refers to the angle vertex she referred in Transcript 2a as the center point where her heels click and, anthropomorphizing the lines, where they all come to meet. As she placed a sticker on her center point, she said, “This is, like right here, the big center point... with all the lines just coming... like this is the place where it's... almost like a center where everybody comes to meet... That's also where my heels click like at the end.” Later, Anna placed two more dot stickers on the mat to mark the tips of her toes (see Figure 3).

Figure 3
Anna's Construction of a System of Auxiliary Lines

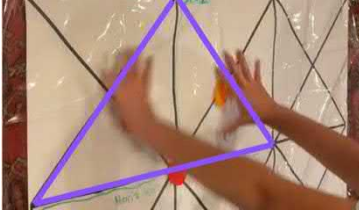


Note. Anna constructed a system of auxiliary lines to overtly show the covert lines she uses to evaluate her dancing posture. The green lines mark the direction of her feet as diagonal to the auxiliary lines. The dot stickers mark the position of her heels (red) and toes (pink and orange). The gray arcs indicate Anna's incarnation of a 90° angle formed by two 45° angles, as in her explanation in Transcript 1b-c.

In Transcript 2 and Figure 3, we see Anna using various material elements in the activity space as semiotic resources to *enhance*, *evaluate*, and *explain* her TSP posture (cf. Abrahamson, 2014). She first acknowledged the existence of the point where her heels met (Transcript 2a), which later she related metaphorically to the red dot as the center of other elements ("a center where everybody comes to meet"). Next, she ensured that her left and right feet, oriented diagonally from the center, constituted angle bisectors of the left and right 90° angles so that the angle *between* her feet became the sum of twice 45° , equal to 90° (confirming Transcript 1b-c).

Next, Anna was asked to perform TSP in different spatial locations, per a standard Balinese choreography sequence. As Transcript 3 will demonstrate, once Anna had adjudicated the GRiD's auxiliary lines as effective evaluation tools for performing TSP, she was now able to "pack" the geometrical construction as a portable system of attentional anchors that she could transport to new locations on the mat, as per the choreographic task of stepwise advancing the TSP form forward.

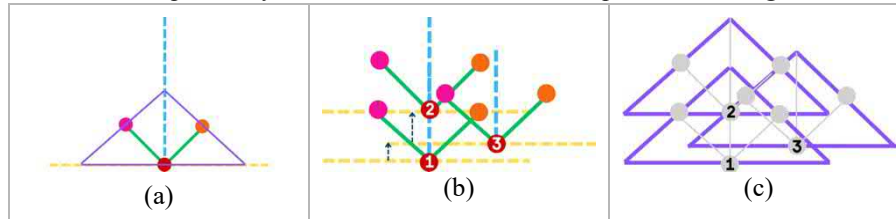
Transcript 3
A Triangle That Always Stays

 (a)	 (b)	 (c)
A: [standing in TSP, heels were on the red dot sticker] when I go like this [right foot steps ahead], I always tend to put my foot like over, like this [left foot steps ahead; she forms TSP a step ahead of the original position; short navy blue arrow traces the direction of movement].	A: But then I keep imagining myself to have another horizontal line here [yellow arrow traces the gesture direction], another vertical line here [blue arrow traces the gesture direction].	A: So I feel like, um this ... [steps back and bends forward] triangle-ish thing [purple lines trace gesture area] always stays with my feet, like whenever I move around.

Confirming the result of Abrahamson and Sánchez-García (2016), when the motor task became more complicated, Anna spontaneously generated an attentional anchor she had not referred to earlier and may have generated on the spur of the moment—she perceived a triangle that draws on the gridlines. Characterizing the triangle as something that *always stays* with her feet, Anna perceived this portable geometric construction as phenomenologically intrinsic to her movement experience. Once this triangle was diagrammed, Anna could elaborate and extend her embodied argument by bringing to bear geometrical reasoning. The conversation continued with another choreographic task that required Anna to move sideways, backward, and forward before

reconfiguring the TSP form. Consistently, Anna used the triangle as a reference structure guiding her movement and posture (see the illustration in Figure 4).

Figure 4
Geometric Properties of Anna's Dance-Oriented Perceptual Reasoning on the GRiD Mat



Note. Anna constructed a triangle (marked in purple) to evaluate her feet's position (a). She then marked her TSP at three different dancing locations in chronological order of 1, 2, and 3 (b); the short vertical black dashed arrows mark the relocations of the horizontal yellow line, which is the base of the purple triangles (in a and c). Anna's triangle at three locations (c).

Anna used available semiotic means (the mat gridlines) to objectify the perceptual construction she had spontaneously generated to enact and evaluate her TSP (q.v., Radford, 2003). In turn, this triangular diagrammatic system became her imaginary means of projecting and calibrating the TSP form as she relocated it across the mat. For an elaboration on the cultural–historical situativity of GRiD's linear matrix, readers are referred to our cognitive-anthropological study of Balinese spatial practices (Apsari & Abrahamson, 2025).

Discussion

Embodiment perspectives conceptualize the core cognitive work in mathematical learning as the explorative process of perceptually coordinating goal-oriented sensorimotor activity (Abrahamson & Sánchez-García, 2016). As such, the traditionally disparate scholarship around mathematics and movement may engage in dialogue that benefits both fields (Abrahamson & Shulman, 2019). In this study, we designed a gridded tarp (GRiD) as a semiotic resource for dancing students to objectify their tacit attentional anchors in the form of geometrical auxiliary constructions. Our empirical data suggest that the study participant indeed alternated between dance and geometry discursive registers grounded in shifting perceptions of the available media (on “keying” registers, see DeLiema et al., 2019).

Our culturally situated activity spontaneously elicited from the participant mathematical reasoning in her attempts to explain her dancing posture (Transcript 1). Once she had voluntarily constructed auxiliary lines, GRiD was presented as bearing potential semiotic resources for her to elaborate on her reasoning. Working with GRiD, Anna could then describe with greater precision the imaginary auxiliary elements she had employed in dancing, thus rendering her implicit routine transparent for others to follow (Transcript 2). For instance, Anna was eager to talk about the auxiliary elements (e.g., center point, lines) that she employed to adjust her feet's position per TSP. Whereas prior studies report that students experience difficulty generating auxiliary lines to solve geometry problems (Foster, 2023), our participant intuitively constructed auxiliary elements that proved effective geometric means of solving her movement task.

By situating geometry discourse in dance settings, our activity created a new *nexus of practice*—a domain of study created by a constellation of different practices connected by various relations (Hui et al., 2017). This nexus allowed Anna and the researcher to establish a mutual understanding of the discourse, enabling them to employ rules, entities, and vocabularies across dance and mathematics without causing confusion. The data show how the designed environment of GRiD both elicited and enabled an individual to objectify their intuitively generated auxiliary lines to bridge the discourses of dance and geometry.

Participating in the designed activity, Anna's dance experience merged with the mathematical universe. Indeed, she became part of the mathematical *model* of her dance strategy (Freudenthal, 1978) and, vice versa, the mathematical model then enhanced her dancing. This ontological merging of geometry and dance constructions as dynamic accessories—her geometric perception of dance—is evident in Anna's utterances, such as “all the lines just come like this” or “this triangle thing always stays with my feet.” Similarly, Anna perceived her dance-oriented attentional anchor as an angle-bisector auxiliary construction running forward from her TSP-formed heels (see Figure 3). Furthermore, Anna's multimodal utterances evidenced that she was blending dance and geometry perceptual registers. For instance, in perceiving the vertex of the angle constructed by her feet configuration (see Transcript 2), Anna was: (1) gesturing—depicting the shape of the vertex as a point; (2) quantifying—describing

the size of the vertex as “this big”; (3) naming—metaphorically referring to the vertex as the “big center point”; and (4) sounding—depicting the vertex as the location where her heels “click.” Finally, as the coordination problem became complex, Anna developed a new geometric attentional anchor (the triangular system, see Figure 4) that integrated the various auxiliary elements. We do not know of previous research demonstrating bilateral reciprocity between mathematical and movement practices.

Conclusion

Marshall McLuhan famously asserted, “The medium is the message.” He argued that the medium we select to convey our message impacts how the meaning of the message is conceived (McLuhan, 1964). In our study, we built a medium, GRiD, that affects how students perceive, communicate, and interact with and about mathematical objects and concepts. Dancers’ tacit perceptual constructions—imaginary points and lines guiding the enactment of their movements—became available for others’ shared scrutiny as indexable semiotic objects. GRiD enables students to render their covert auxiliary construction as overt diagrammatic constructions. The medium we designed offers material anchors for blending dance and mathematics discourses with low-cost technology affordable for students in remote areas.

The designed learning environment served as a field of promoted action (Abrahamson & Trninic, 2015; Reed & Bril, 1996) occasioning productive opportunities to talk about mathematics when dancing and to talk about dance when mathematizing. Thus, the merged dance/geometry activities elicited, encouraged, and legitimized multimodal and multimedia forms of mathematical reasoning and discourse that included whole-body movement, gesture, speech, and drawing. Dancing geometry or, more technically, blending dance and geometry discourses, enables students to experience complex mathematics as practical situated forms of reasoning. More broadly, situating students’ mathematical experiences in personally meaningful cultural heritage activities could render schooling more playful and creative (Thompson, 2022).

References

- Abrahamson, D. (2014). Building educational activities for understanding: An elaboration on the embodied-design framework and its epistemic grounds. *International Journal of Child-Computer Interaction*, 2(1), 1–16. <https://doi.org/10.1016/j.ijcci.2014.07.002>
- Abrahamson, D., Nathan, M. J., Williams-Pierce, C., Walkington, C., Ottmar, E. R., Soto, H., & Alibali, M. W. (2020). The future of embodied design for mathematics teaching and learning. *Frontiers in Education*, 5, 147. <https://doi.org/10.3389/educ.2020.00147>
- Abrahamson, D., & Sánchez-García, R. (2016). Learning is moving in new ways: The ecological dynamics of mathematics education. *Journal of the Learning Sciences*, 25, 203–239. <https://doi.org/10.1080/10508406.2016.1143370>
- Abrahamson, D., & Shulman, A. (2019). Co-constructing movement in mathematics and dance: An interdisciplinary pedagogical dialogue on subjectivity and awareness. *Feldenkrais Research Journal*, 6, 1–24.
- Abrahamson, D., & Trninic, D. (2015). Bringing forth mathematical concepts: Signifying sensorimotor enactment in fields of promoted action. *ZDM—Mathematics Education*, 47, 295–306. <https://doi.org/10.1007/s11858-014-0620-0>
- Apsari, R. A., & Abrahamson, D. (2024). Dancing geometry: Imagining auxiliary lines by reflecting on physical movement. *International Journal of Mathematical Education in Science and Technology*. <https://doi.org/10.1080/0020739X.2024.2427099>
- Apsari, R. A., & Abrahamson, D. (2025). Culturally situated embodied design: A cognitive-anthropological investigation of Balinese spatial practice toward grounding geometry. Submitted for publication in the *Proceedings of the 47th Annual Meeting of PME-NA*. (Manuscript under review).
- Ba, H., & Abrahamson, D. (2021). Taking design to task: A dialogue on task-initiation in STEM activities. *Educational Designer*, 4(14). <http://www.educationaldesigner.org/ed/volume4/issue14/article54/>
- Clark, J. O., & Ando, T. (2014). Geometry, embodied cognition and choreographic praxis. *International Journal of Performance Arts and Digital Media*, 10(2), 179–192. <https://doi.org/10.1080/14794713.2014.946285>
- DeLiema, D., Enyedy, N., & Danish, J. A. (2019). Roles, rules, and keys: How different play configurations shape collaborative science inquiry. *Journal of the Learning Sciences*, 28(4–5), 513–555. <https://doi.org/10.1080/10508406.2019.1675071>
- Doorman, M., Van den Heuvel-Panhuizen, M., & Goddijn, A. (2020). The emergence of meaningful geometry. In M. Van den Heuvel-Panhuizen (Ed.), *National reflections on the Netherlands didactics of mathematics: Teaching and learning in the context of realistic mathematics education*. Springer International Publishing. <https://doi.org/10.1007/978-3-030-33824-4>

- Foster, C. (2023). Problem solving in the mathematics curriculum: From domain-general strategies to domain-specific tactics. *The Curriculum Journal*, *curj.213*. <https://doi.org/10.1002/curj.213>
- Freudenthal, H. (1978). *Weeding and sowing: Preface to a science of mathematical education*. Springer Dordrecht. <https://doi.org/10.1007/0-306-47234-1>
- Gerofsky, S. (2011). Seeing the graph vs. Being the graph: Gesture, engagement and awareness in school mathematics. In G. Stam & M. Ishino (Eds.), *Integrating gestures: The interdisciplinary nature of gesture* (Vol. 4, pp. 245–256). John Benjamins Publishing Company. <https://doi.org/10.1075/gs.4.22ger>
- Hui, A., Schatzki, T. R., & Shove, E. (2017). *The nexus of practices: Connections, constellations, practitioners* (A. Hui, T. R. Schatzki, & E. Shove, Eds.). Routledge. <https://doi.org/10.4324/9781315560816>
- Hutto, D. D., & Sánchez-García, R. (2015). Choking RECTified: Embodied expertise beyond Dreyfus. *Phenomenology and the Cognitive Sciences*, *14*, 309–331. <https://doi.org/10.1007/S11097-014-9380-0>
- Johnson-Glenberg, M. C. (2022). Evaluating embodied immersive STEM VR using the quality of education in virtual reality rubric (QUIVRR). In S. L. Macrine & J. M. B. Fugate (Eds.), *Movement matters: How embodied cognition informs teaching and learning* (pp. 237–258). The MIT Press. <https://doi.org/10.7551/mitpress/13593.003.0023>
- Kirsh, D. (2010). Thinking with external representations. *AI & Society*, *25*(4), 441–454. <https://doi.org/10.1007/s00146-010-0272-8>
- McLuhan, M. (1964). *Understanding media: The extensions of man* (1st ed.). McGraw-Hill.
- Mondada, L. (2019). Contemporary issues in conversation analysis: Embodiment and materiality, multimodality and multisensoriality in social interaction. *Journal of Pragmatics*, *145*, 47–62. <https://doi.org/10.1016/j.pragma.2019.01.016>
- OECD. (2022). *Mending the education divide: Getting strong teachers to the schools that need them most*. OECD. <https://doi.org/10.1787/92b75874-en>
- Palatnik, A., Abrahamson, D., Baccaglioni-Frank, A., Ng, O. L., Shvarts, A., & Swidan, O. (2023). Theory and practice of designing embodied mathematics learning [Research forum]. In M. Ayalon, B. Koichu, R. Leikin, L. Rubel, & M. Tabach (Eds.), *Proceedings of the 46th Conference of the International Group for the Psychology of Mathematics Education* (Vol. 1). PME 46.
- Palatnik, A., & Sigler, A. (2018). Focusing attention on auxiliary lines when introduced into geometric problems. *International Journal of Mathematical Education in Science and Technology*, *50*(2), 202–215. <https://doi.org/10.1080/0020739X.2018.1489076>
- Radford, L. (2003). Gestures, speech, and the sprouting of signs: A semiotic-cultural approach to students' types of generalization. *Mathematical Thinking and Learning*, *5*(1), 37–70. https://doi.org/10.1207/S15327833MTL0501_02
- Reed, E. S., & Bril, B. (1996). The primacy of action in development. In M. L. Latash & M. T. Turvey (Eds.), *Dexterity and its development* (pp. 431–451). Lawrence Erlbaum Associates.
- Shapiro, L., & Stolz, S. A. (2019). Embodied cognition and its significance for education. *Theory and Research in Education*, *17*(1), 19–39. <https://doi.org/10.1177/1477878518822149>
- Siegler, R. S. (2006). Microgenetic analyses of learning. In D. Kuhn & R. Siegler (Eds.), *Handbook of child psychology* (6th ed., Vols. 2, Cognition, perception, and language, pp. 464–510). Wiley.
- Skemp, R. (1976). Relational understanding and instrumental understanding. *Mathematics Teaching*, *77*, 20–26.
- Soto, H. (2022). Acting out Euclidean transformations. *PRIMUS*, *32*(8), 902–916. <https://doi.org/10.1080/10511970.2021.1954115>
- Steffe, L. P., & Thompson, P. W. (2000). Teaching experiment methodology: Underlying principles and essential elements. In R. Lesh & A. E. Kelly (Eds.), *Research design in mathematics and science education* (pp. 267–307). Erlbaum.
- Stylianides, G. J. (2008). An analytic framework of reasoning and proof. *For the Learning of Mathematics*, *28*(1), 9–16.
- Tancredi, S., Chen, R. S. Y., Krause, C., Abrahamson, D., & Gomez, F. (2021). Getting up to SpEED: Special Education Embodied Design for sensorially equitable inclusion. *Education Sciences and Society*, *1*, 114–136. <https://doi.org/10.3280/ess1-2021oa11818>
- Thompson, N. (2022). “Math cleverly disguised as/with string”: Experienced weavers’ engagement with mathematics. *Research in Mathematics Education*, 1–21. <https://doi.org/10.1080/14794802.2022.2133005>
- Vogelstein, L., Brady, C., & Hall, R. (2019). Reenacting mathematical concepts found in large-scale dance performance can provide both material and method for ensemble learning. *ZDM—Mathematics Education*, *51*(2), 331–346. <https://doi.org/10.1007/s11858-019-01030-2>